Web services: WMS, WFS, WCS

geo0920 · maps and data from a server

example provider: Geobasis NRW



Download a file or ask a server?

	download a file	ask a web service
the data live …	on your disk	on the provider’s server
you get …	the whole tile or dataset	only your extent and scale
up to date?	as of the download day	always the current version
offline?	yes	no: needs internet and the server
good for …	analysis, editing, archiving	background maps, a quick look, small queries

Same open data both ways: Geobasis NRW offers files (opengeodata.nrw.de) and services (wms / wfs / wcs.nrw.de).

Text and tables: own work, CC BY-SA 4.0. Service list: Geobasis NRW, Geodatendienste (2026).

OGC web services at a glance

service	you get back	use it to …	Geobasis NRW examples
WMS **Web Map Service**	a picture (PNG, JPEG) drawn for your extent	look: background maps	orthophotos (DOP), topographic maps (DTK), hillshade
WMTS **Web Map Tile Service**	ready-made picture tiles	look, fast	DOP, DTK collection
WFS / OGC API Features	vector features with attributes	select, count, join	administrative boundaries, buildings
WCS **Web Coverage Service**	raster values, e.g. elevation in m	calculate	terrain model DGM1, surface model DOM1

Wikipedia: Open Geospatial Consortium · Web Map Service · Web Map Tile Service · Web Feature Service · Web Coverage Service

QGIS manual: 19 Working with OGC / ISO protocols

Text and tables: own work, CC BY-SA 4.0. Service list: Geobasis NRW, Geodatendienste (2026).



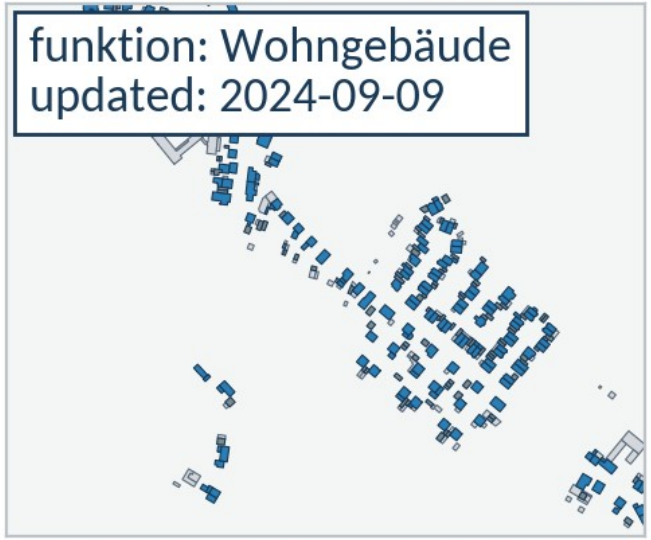
Three services, one place

WMS: a picture



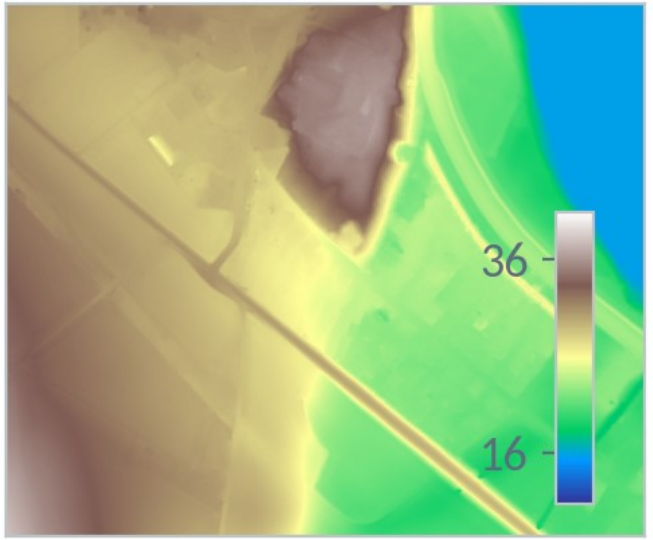
orthophoto: colours

WFS: objects + attributes



395 buildings (ALKIS)

WCS: values



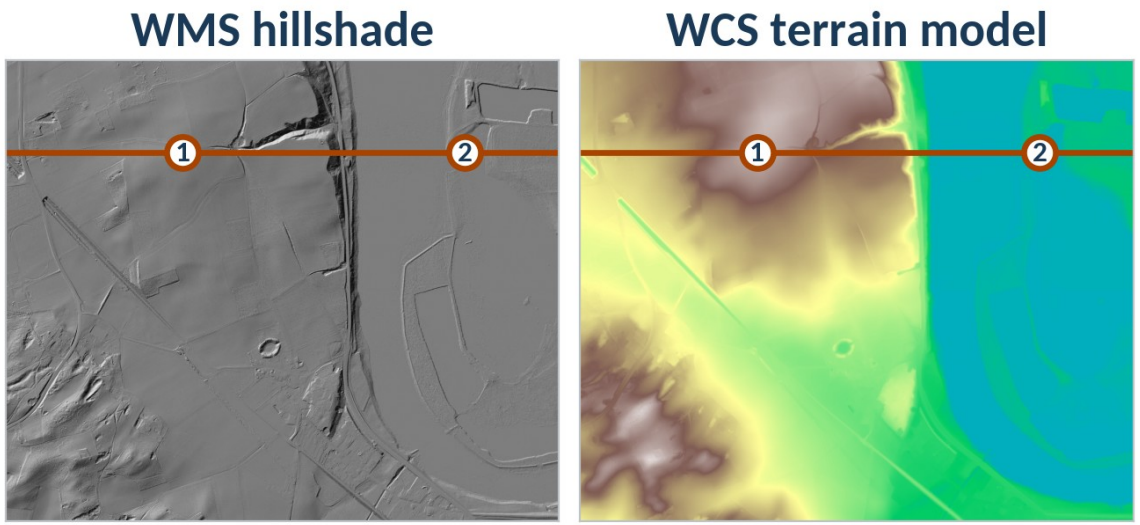
terrain model: metres

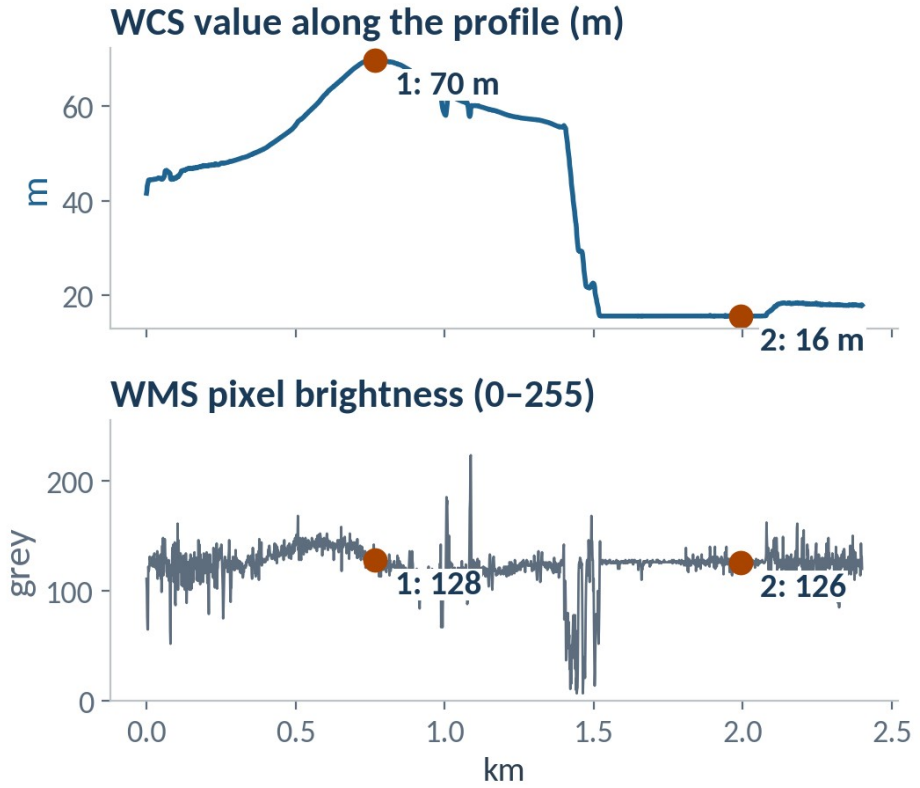
residential · garage, parking · farm building · other

QGIS manual: 19.1 WMS/WMTS · 19.2 WCS · 19.3 WFS

Figures: own work, CC BY-SA 4.0. Data: © Geobasis NRW (2026), dl-de/zero-2.0, via WMS, WFS, WCS.

A picture of the terrain or the terrain itself?





WMS hillshade: brightness 0–255

WCS terrain model: metres, to calculate with

Wikipedia: Terrain cartography: shaded relief · Digital elevation model

QGIS manual: 19.1 WMS/WMTS · 19.2 WCS

Figures: own work, CC BY-SA 4.0. Data: © Geobasis NRW (2026), dl-de/zero-2.0, via WMS and WCS.

Why QGIS freezes: data volume

what you switch on	2 km × 2 km	Kreis Wesel (40 × 40 km)	NRW
WMS hillshade (picture)	2.3 MB · 2 s	1.9 MB · 2 s	1.5 MB · 4 s
WCS terrain model, screen-sized	4.5 MB · 3 s	5.9 MB · 5 s	refused
WCS terrain model, full 1 m	1 request · 8 MB	≥ 260 requests · 2 GB	≥ 8 500 · 60 GB
WFS buildings (ALKIS)	1 108 · 1 MB · 2 s	851 000 · ≈ 0.8 GB	12.5 million · ≈ 12 GB

WMS: cost = your screen. **WCS:** cost = area the server reads. **WFS:** cost = number of objects, all in one answer.

Zoom in first – then switch WFS or WCS layers on.

QGIS manual: 19.2 WCS client · 19.3 WFS client · Options ▸ Network

Measured 2026-10-11 against the Geobasis NRW services; screen 1500 × 1500 px; WFS for Wesel and NRW counted, not downloaded.



Fetch once, save locally

service	Save As gives you …	offline good for	watch out
WMS / WMTS	a picture (GeoTIFF)	background map	size grows with the resolution
WFS	features + attributes (GeoPackage)	select, count, join	only what is loaded for the extent
WCS	values, e.g. metres (GeoTIFF)	calculate	≤ 2 km × 2 km per request at 1 m

In QGIS

Layers panel ▸ right-click the layer ▸ **Export ▸ Save As…**

Extent: **Map Canvas Extent** ▸ format GeoTIFF / GeoPackage

The cache is not a copy (WMS: 24 h). A saved file is a snapshot: note the download date.

QGIS manual: 19.2 WCS client · 19.3 WFS client · Options ▸ Network (cache)

Text: own work, CC BY-SA 4.0.

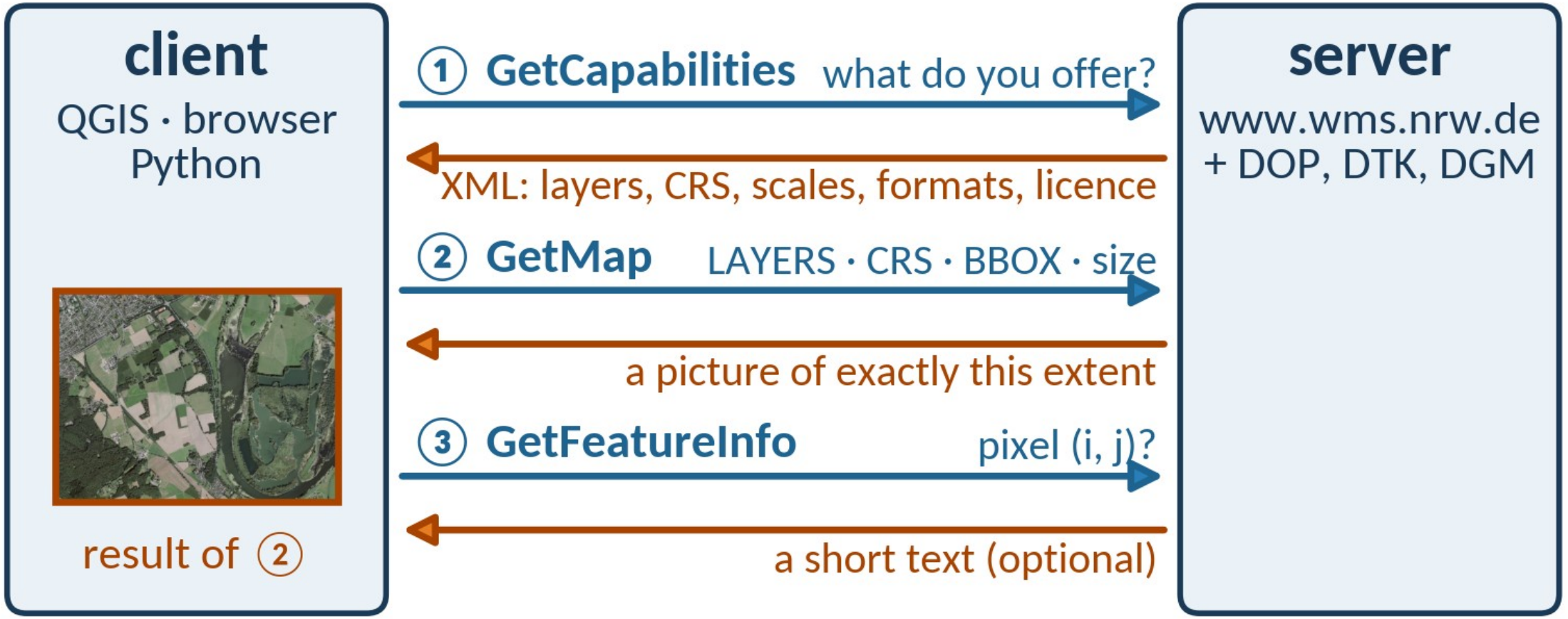


Web Map Service (WMS)

GetCapabilities · GetMap · background maps



How a WMS works



Wikipedia: Web Map Service

QGIS manual: 19.1 WMS/WMTS client

Figures: own work, CC BY-SA 4.0. Maps: © Geobasis NRW (2026), dl-de/zero-2.0, via WMS.

Three background maps from one provider

orthophoto · DOP



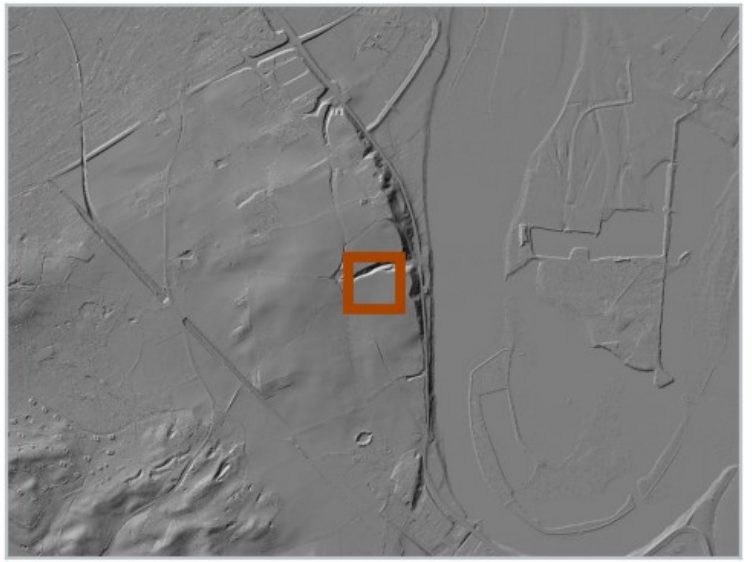
`nw_dop_rgb`

topographic map · DTK



`nw_dtk_col`

hillshade · DGM



`nw_dgm-schummerung_pan`

Wikipedia: Orthophoto · Topographic map · Terrain cartography: shaded relief · Vetera

Figures: own work, CC BY-SA 4.0. Maps: © Geobasis NRW (2026), dl-de/zero-2.0, via WMS. Area: Fürstenberg, Xanten-Birten, 4 km × 3 km.

GetCapabilities: what does the server offer?

```
…/geobasis/wms_nw_dop?SERVICE=WMS
   &REQUEST=GetCapabilities
```

```
<Service>
  <Title>WMS NW DOP</Title>
  <Fees>… Datenlizenz Deutschland – Zero
</Service>
<Capability> <Layer> …
  <Layer queryable="0">
    <Name>nw_dop_rgb</Name>
    <Title>DOP Farbe</Title>
    <CRS>EPSG:25832</CRS> … (20 CRSs)
    <BoundingBox CRS="EPSG:25832"
       minx="270100" miny="5.55e+06" …/>
```

- **Name** = what the request needs; **Title** = what QGIS shows
- **CRS**: the server reprojects for you; ask for **EPSG:25832** in NRW
- **BoundingBox**: where data exist
- **Min/MaxScale**: outside, the layer stays empty
- **Fees**: the licence

Excerpt from the live capabilities of wms_nw_dop (2026-10-11), shortened.



GetMap: a map is just a URL

parameter	value
SERVICE · VERSION	WMS · 1.3.0
REQUEST	GetMap
LAYERS	nw_dop_rgb
CRS	EPSG:25832
BBOX	323200,5723100, 327200,5726100
WIDTH · HEIGHT	1200 · 900 (pixels)
FORMAT	image/png



BBOX = min E, min N, max E, max N

3.3 m per pixel ≈ 1 : 12 000

Maps: © Geobasis NRW (2026), dl-de/zero-2.0, via WMS.

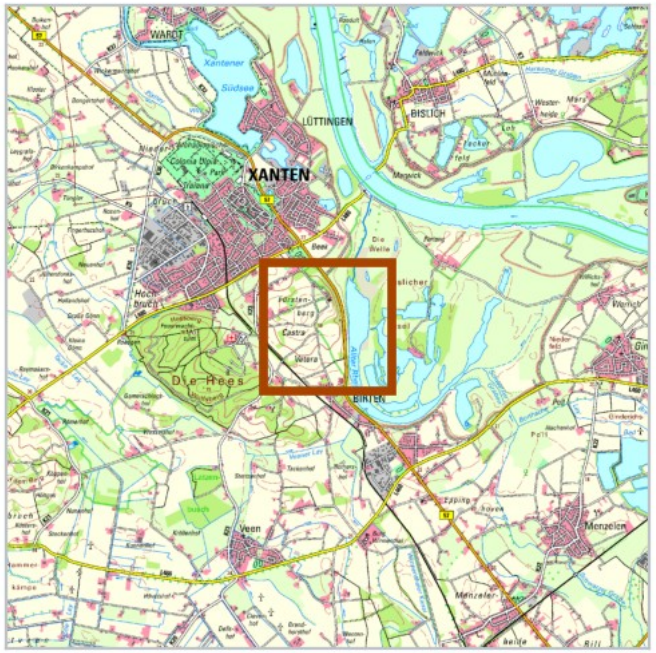
One layer, many maps: the DTK collection

1 : 8 000



→ P10 NRW

1 : 40 000



→ DTK50

1 : 200 000



→ DTK250 (BKG)

The server picks the map series for your scale. **Small scales: name the BKG source.**

QGIS manual: 19.1.5 Tilesets (WMTS)

Figures: own work, CC BY-SA 4.0. Maps: © Geobasis NRW (2026), dl-de/zero-2.0, via WMS. 1 : 200 000: © GeoBasis-DE / BKG (2026), dl-de/by-2-0.

Geobasis NRW services in QGIS

In QGIS

1 Geobasis NRW ▸ Webdienste ▸ Geodatendienste ▸ Downloads: **“Dateien zur Befüllung von Auswahllisten in QGIS”** (ZIP), unzip

2 **Layer ▸ Data Source Manager ▸ WMS/WMTS ▸ Load** → nw_wms_geodatendienste.xml → tick the three services

3 pick a service in the dropdown ▸ **Connect** ▸ select a layer ▸ **Add**

4 Project ▸ Properties ▸ CRS: **EPSG:25832**

tick these three, then add

- **… Orthophotos (DOP) [WMS]**
  → DOP Farbe
- **… DTK Sammeldienst [WMS]**
  → DTK Farbe
- **… Schummerung (DGM-Schummerung)**
  → … Graustufen NW

Geobasis NRW Webdienste

▸ Geodatendienste ▸ ZIP

With screenshots: Anleitung WMS, ch. 4

QGIS manual: 19.1.3 Selecting WMS/WMTS servers · 19.1.4 Loading WMS/WMTS layers

Service URLs and layer names: Geobasis NRW capabilities (2026-10-11).

Pitfalls

you see …	because …	do this
an empty layer	outside the scale range or outside NRW	zoom in; check Min/MaxScale
the map is shifted or rotated	EPSG:4326 in 1.3.0 is lat, lon	ask for EPSG:25832
Identify gives no elevation	a WMS sends a picture, not data	WCS for values, WFS for features
slow panning	every move = new GetMap	WMTS, or download the data
a source line is missing	small DTK scales are BKG data	© GeoBasis-DE / BKG, dl-de/by-2-0

QGIS manual: 19.1.6 Using the Identify tool · Options ▸ Network (cache, timeout)

Text: own work, CC BY-SA 4.0.



Geobasis NRW: guides and service lists

Start here: Geobasis NRW, Webdienste (all web services, guides, licence)

Service URLs: Geodatendienste · QGIS connection lists: ZIP (“Dateien zur Befüllung von Auswahllisten in QGIS”, under Downloads)

Guides (German): WMS · WFS · WCS – requests explained, QGIS chapter with the XML import

QGIS manual: 19 Working with OGC / ISO protocols (latest release)

Standard: OGC Web Map Service 1.3.0

Licence: Datenlizenz Deutschland – Zero 2.0 (Geobasis NRW) · Namensnennung 2.0 (BKG)

Links checked 2026-10-11.



Sources and further reading

OGC web services at a glance: Open Geospatial Consortium · Web Map Service · Web Map Tile Service · Web Feature Service · Web Coverage Service · 19 Working with OGC / ISO protocols

Three services, one place: 19.1 WMS/WMTS · 19.2 WCS · 19.3 WFS

A picture of the terrain or the terrain itself?: Terrain cartography: shaded relief · Digital elevation model · 19.1 WMS/WMTS · 19.2 WCS

Why QGIS freezes: data volume: 19.2 WCS client · 19.3 WFS client · Options ▸ Network

Fetch once, save locally: 19.2 WCS client · 19.3 WFS client · Options ▸ Network (cache)

How a WMS works: Web Map Service · 19.1 WMS/WMTS client

Three background maps from one provider: Orthophoto · Topographic map · Terrain cartography: shaded relief · Vetera

One layer, many maps: the DTK collection: 19.1.5 Tilesets (WMTS)

Geobasis NRW services in QGIS: 19.1.3 Selecting WMS/WMTS servers · 19.1.4 Loading WMS/WMTS layers

Pitfalls: 19.1.6 Using the Identify tool · Options ▸ Network (cache, timeout)

Figures: own work (make_slides.py), CC BY-SA 4.0; map pictures © Geobasis NRW, dl-de/zero-2.0 (small scales © GeoBasis-DE / BKG, dl-de/by-2-0).

